

Topic: Multilingual Mobile App Review Dataset.

Phase - 1

Objective:

To analyse mobile app reviews and ratings to understand user satisfaction, app performance, and review trends.

Key Objectives

1. Clean and prepare the review data by handling missing values, inconsistencies.
2. Analyse user ratings to identify overall satisfaction and the distribution of positive and negative ratings.
3. Compare app performance based on the number of reviews, average ratings, and helpful votes.
4. Identify trends over time by analysing monthly and yearly review patterns.
5. Understand user behaviour based on language, country, gender, and device type.

About Dataset

This comprehensive synthetic dataset contains 2,514 authentic mobile app reviews spanning 40+ popular applications across 24 different languages.

❖ Language Distribution

The dataset includes reviews in 24 languages:

- European: English (en), Spanish (es), French (fr), German (de), Italian (it), Russian (ru), Polish (pl), Dutch (nl), Swedish (sv), Danish (da), Norwegian (no), Finnish (fi), Portuguese (pt)
- Asian: Chinese (zh), Hindi (hi), Japanese (ja), Korean (ko), Thai (th), Vietnamese (vi), Indonesian (id), Malay (ms)
- Other: Arabic (ar), Turkish (tr), Filipino (tl)

❖ Application Categories

Reviews cover 18 distinct categories:

- Social Networking
- Entertainment
- Productivity
- Travel & Local
- Music & Audio
- Video Players & Editors
- Shopping
- Navigation

- Finance
- Communication
- Education
- Photography
- Dating
- Business
- Utilities
- Health & Fitness
- Games
- News & Magazines

❖ Popular Apps Included

40+ applications including:

- Social: WhatsApp, Instagram, Facebook, Snapchat, TikTok, LinkedIn, Twitter, Reddit, Pinterest
- Entertainment: YouTube, Netflix, Spotify
- Productivity: Microsoft Office, Google Drive, Dropbox, OneDrive, Zoom, Discord
- Travel: Uber, Lyft, Airbnb, Booking.com, Google Maps, Waze
- Finance: PayPal, Venmo
- Education: Duolingo, Khan Academy, Coursera, Udemy
- Tools: Grammarly, Canva, Adobe Photoshop, VLC, MX Player

❖ Geographic Distribution

Reviews from 24 countries across all continents:

- Asia: China, India, Japan, South Korea, Thailand, Vietnam, Indonesia, Malaysia, Philippines, Pakistan, Bangladesh
- Europe: Germany, United Kingdom, France, Italy, Spain, Russia, Turkey, Poland
- Americas: United States, Canada, Brazil, Mexico
- Oceania: Australia
- Africa: Nigeria

Column Specifications

Column Name	Data Type	Description	Sample Values	Null Count
review_id	Integer	Unique identifier for each review	1, 2, 3, ...	0
user_id	String*	User identifier (should be integer)	"1967825", "9242600"	0
app_name	String	Name of the mobile application	WhatsApp, Instagram, TikTok	0
app_category	String	Application category	Social Networking, Entertainment	0
review_text	String	Multilingual review content	"This app is amazing!"	63
review_language	String	ISO language code	en, es, fr, zh, hi, ar	0
Rating	Mixed*	App rating (1.0-5.0, some as strings)	4.5, "3.2", 1.1	38
review_date	DateTime	Timestamp of review submission	2024-10-09 19:26:40	0
verified_purchase	Boolean	Purchase verification status	True, False	0
device_type	String	Device platform	Android, iOS, iPad, Windows Phone	0
num_helpful_votes	Mixed*	Helpfulness votes (some as strings)	65, "209", 163	0
user_age	Float*	User age (should be integer)	14.0, 18.0, 67.0	0
user_country	String	User's country	China, Germany, Nigeria	50
user_gender	String	User gender	Male, Female, Non-binary, Prefer not to say	88
app_version	String	Application version number	1.4, v8.9, 2.8.37.5926	25

Data Cleaning & Preprocessing Steps

1. Identify Missing Values

Checked all columns for blank or missing values using COUNTBLANK() function

- Review Text → Missing values identified. Null values are replaced with text “No Review” using Find & Replace
- Rating → Missing values identified and null values are replaced with mean value
- Country → Blanks replaced with Unknown
- Gender → Blanks replaced with Unknown
- App Version → Blanks replaced with Unknown

2. Remove Inconsistencies

- Removing inconsistencies in Review language column , values are filled by Find & Replace
For ex: Replaced “en” with “English”

3. Remove Duplicates

Method — Conditional Formatting ★

1. Select the review_id column (excluding the header).
2. Go to Home → Conditional Formatting.
3. Select Highlight Cells Rules → Duplicate Values.
4. Keep Duplicate selected.
5. Choose a highlight color.
6. Click OK.

- There is no duplicate records in the dataset

4.Changing the data type

The data types of the columns were reviewed and corrected according to their purpose. Numeric fields such as Rating, User Age, and Helpful Votes were set to numeric data types, Review Date was converted to Date/Time format, and categorical fields such as App Name, Language, Country, Gender, and App Version were stored as Text.

5.Data Transformation

New columns, Review Date Year and Review Date Month, were created from the review_date column using the TEXT() function.

=TEXT([@[Review_date]],"YYYY")

=TEXT([@[Review_date]],"MMMM")

Pivot Tables & Charts

1.Reviews by App

2.Reviews by App Category

- 3.Reviews by Language
- 4.Reviwes by Country
- 6.Reviews by Gender
- 7.Reviews by Device Type
- 8.Verfied vs Non Verified Purchase
- 9.Monthly Review
- 10.Yearly Review
- 11.Rating Distribution

Conclusion

The analysis shows moderate overall user satisfaction, with an average rating of 3.02. Review activity is strongest in 2024, while the large number of helpful votes indicates high user engagement. The rating distribution also highlights opportunities for apps to improve user satisfaction.

Phase – 2

After the phase 1(cleaning and transformation process), the cleaned dataset was loaded into **Power BI** for further analysis and visualization.

In phase 2, developed an interactive **Power BI dashboard** to analyse multilingual mobile app reviews and generate actionable insights into app performance, user ratings, review behaviour, helpful votes, verified purchases, and user demographics with the help of DAX calculated columns and measures.

DAX Calculations

Several measures and calculated column were created to support the dashboard.

Calculated Column – Rating Category

A calculated column was created to classify numerical ratings into meaningful categories:

- **Poor** – Rating below 2
- **Average** – Rating from 2 to below 3
- **Good** – Rating from 3 to below 4.5
- **Excellent** – Rating of 4.5 or above

This categorization makes it easier to analyse the overall distribution of app ratings rather than relying only on individual rating values.

Key Measures

The following measures were created:

Measure	Purpose
Total Reviews	Measures the total number of reviews
Average Rating	Calculates the overall average user rating
Total Helpful Votes	Measures the total number of helpful votes received
Verified Purchase %	Calculates the percentage of reviews from verified purchases
Number of Apps	Counts the unique apps in the dataset
Average User Age	Calculates the average age of users
Selected App	Dynamically displays the currently selected app or "All Apps"

These measures make the dashboard **dynamic and interactive**, allowing the KPIs and visualizations to respond to user selections.

Slicers

- App Name
- App Category
- User Language
- User Country
- Verified Purchase
- Date

Power BI Dashboard Development

The final Power BI report was designed with **five analytical pages**, each having a specific purpose by page navigation.

Page 1 – Executive Analysis

The Executive Analysis page provides a **high-level overview of the entire dataset**.

It focuses on important KPIs such as:

- Total Reviews
- Average Rating
- Total Helpful Votes
- Verified Purchase %
- Number of Apps
- Average User Age

Charts and visuals provide an overall understanding of:

- Reviews by app – Clustered bar chart
- Reviews by category-Clustered bar chart
- Reviews over month-Area Chart

This page is designed for users who want to understand the **overall performance of the mobile applications at a glance**.

Page 2 – App & Rating Analysis

This page focuses specifically on **app performance and rating**.

Charts and visuals provide an overall understanding of:

- Average rating by app – Clustered bar chart
- Average rating by app category – Clustered bar chart
- Reviews by app category – Clustered bar chart
- Rating distribution – Pie chart
- Total helpful votes by app – Clustered bar chart

Page 3 – User & Review Analysis

This page examines the **characteristics and behaviour of users who submitted reviews**.

The analysis includes areas such as:

- Average User age – Measure
- Reviews by language – Clustered column chart
- Reviews by country – Map
- Reviews by device type – Donut chart
- Reviews by gender – Pie chart

Page 4 – App Details / Drill-Through

A dedicated **App Details** page was created using Power BI's **Drill-through functionality**.

This allows users to select a particular app from another page and navigate to a detailed view of that app.

The page dynamically displays information related to the selected application.

The **Selected App** measure provides a dynamic title such as: Selected Name from other pages.

Ex:"Twitter"

depending on the user's selection. The page includes KPI cards and different charts

Page 5 – Insights & Recommendations

The final page summarizes the major findings from the analysis and converts the analytical results into **actionable insights and recommendations**.

The page can be used by stakeholders to understand:

- Which apps perform strongly
- Which apps require improvement
- Rating trends
- User review behaviour
- Verified purchase patterns
- Customer engagement through helpful votes
- Areas requiring further investigation

This page bridges the gap between **data analysis and business decision-making**.



Interactive Features

Several interactive Power BI features were incorporated into the dashboard.

◆ **Slicers**

Slicers allow users to dynamically filter the dashboard based on relevant fields such as:

- App
- Rating
- Rating Category
- Other available dimensions

This enables users to perform their own analysis without modifying the report.

◆ **Cross-Filtering**

The charts were configured to interact with each other.

For example:

Selecting a particular rating category in one visual can dynamically filter related charts and KPI cards.

This allows users to explore relationships within the dataset.

◆ **Drill-Through**

The App Details page uses **Drill-through** to provide a detailed analysis of an individual application.

Reset Page Button

A **Reset Page** button was added to restore the dashboard to its normal/default state.

If a user selects multiple charts or applies different visual interactions, clicking the reset button returns the page to the predefined dashboard view.

Reset Filters Button

A separate **Reset Filters** button was implemented to reset the slicers to their default state.

This gives users two different controls:

Reset Page → restores the complete page state.

Reset Filters → clears the applied slicer filters.

Conclusion:

Developed an interactive Power BI dashboard to analyze multilingual mobile application reviews and uncover insights into app performance, user ratings, review behaviour, helpful votes, verified purchases, and user demographics.

The project involved comprehensive data cleaning and transformation followed by the creation of calculated columns and DAX measures for KPI-driven analysis. A five-page dashboard was developed covering Executive Analysis, App & Rating Analysis, User & Review Analysis, App Details with Drill-through functionality, and Insights & Recommendations.

Interactive slicers, cross-filtering, dynamic titles, drill-through navigation, and dedicated Reset Page and Reset Filters buttons were incorporated to enhance usability. The final solution transforms raw review data into an intuitive business intelligence dashboard that supports comparative analysis, identification of high- and low-performing applications, and data-driven recommendations for improving app performance and user experience.